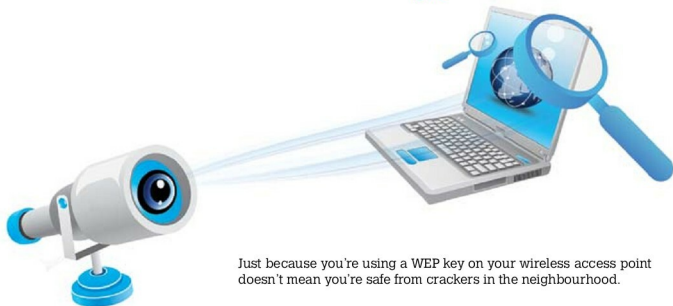


It's So Easy to See Your Network Activity, hah!



Just because you're using a WEP key on your wireless access point doesn't mean you're safe from crackers in the neighbourhood.

A sniffer is basically a network analyser. Likewise, a wireless sniffer is software that can analyse the traffic over a wireless network. The data thus obtained can be used for various purposes—debugging network problems, for instance. These tools can also grab all the non-encrypted data from the network, and hence can be used to crack unsecured networks. This is one of the major reasons why sniffers are a threat to networks.

Detecting the presence of such sniffers is a challenge in itself. On the other hand, you can use these tools to analyse your own networks and check the extent to which they are secure against threats. You could say that the sniffers give you an X-ray view of your network.

Sniffers provide real-time packet data from local, as well as remote machines. Some network analysers even have the ability to alert you of potential developing problems, or bottlenecks that are occurring in real-time. Some have the capability of capturing packet streams and allow you to view these packet streams and edit them.

There are many such sniffing software available on Linux, UNIX, BSD, Windows, etc. Most of the commercial software is quite costly. That, and the fact that I hate Windows, means I will be using one of the popular free software under Linux for sniffing wireless networks and to crack a WEP protected network.

This article is only for educational purposes and I will

be demonstrating the use of sniffers by trying to crack my own wireless network. I will not be liable for any criminal act committed by the reader.

Basic networking information

You will need to know some basics of computer networking in order to fully understand the working of a sniffer tool. Every network device has a MAC (Media Access Control) address. Let's consider a wireless network and, say, four different wireless network cards in its proximity that are connected to that network. The wireless network simultaneously transmits data for all four cards (four computers with wireless networks). Data for each network card is recognised by the MAC address of the corresponding network card. Generally, a network card only receives the data designated for its MAC address. However, when a card is put into what is known as a 'promiscuous mode', it will look at all of the packets being transmitted by the wireless network.

Wireless networks are not the same as cable networks. All computers can access all the data, but generally, they ignore all available data except for the ones designated for them. However, they no longer ignore the data when in 'promiscuous mode', which is the basic feature of sniffing.

There are mainly two methods to achieve this. One is where you connect to the WAP (wireless access point) using your computer to receive all the traffic transmitted